

EduGenie

main.py — FastAPI App

Main FastAPI application. It connects the six project functions to REST endpoints: /qa, /explain, /quiz, /summarize and /learn/recommendations.

Complete Python Code

```
from fastapi import FastAPI
from pydantic import BaseModel
from qna import answer_question
from explanation_module import explain_topic
from quiz_module import generate_quiz
from summary_module import summarize_text
from learning_path import get_learning_recommendations

app = FastAPI(title="EduGenie")

class TextRequest(BaseModel):
    text: str

class QuestionRequest(BaseModel):
    question: str

class LearningRequest(BaseModel):
    topic: str
    level: str = "beginner"

@app.get("/")
def home():
    return {"message": "Welcome to EduGenie"}

@app.post("/qa")
def qa(r: QuestionRequest):
    return {"answer": answer_question(r.question)}

@app.post("/explain")
def explain(r: TextRequest):
    return {"explanation": explain_topic(r.text)}

@app.post("/quiz")
def quiz(r: TextRequest):
    return {"quiz": generate_quiz(r.text)}

@app.post("/summarize")
def summarize(r: TextRequest):
    return {"summary": summarize_text(r.text)}

@app.post("/learn/recommendations")
def recommendations(r: LearningRequest):
    return {"recommendations": get_learning_recommendations(r.topic, r.level)}

# Run: uvicorn main:app --reload
```